

Lattice Point Counting in Arithmetic Statistics

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Remark. Since this work is not to be formally published, I freely copy-and-pasted some text from other papers (mostly my own, but also [?]) when explaining work that appears there. I also used ChatGPT to produce the diagrams.

1 Crash course and an introduction to the projects

Today, we will introduce only one question in arithmetic statistics; namely, that of *cubic field counting*. Let $N_3^\pm(X)$ denote the number of isomorphism classes of cubic fields F satisfying $0 < \pm \text{Disc}(F) < X$. Then, we have the following:

Theorem 1 *We have*

$$(1) \quad N_3^\pm(X) = C^\pm \frac{1}{12\zeta(3)} X + K^\pm \frac{4\zeta(1/3)}{5\Gamma(2/3)^3\zeta(5/3)} X^{5/6} + O(X^{2/3}(\log X)^{2.09}),$$

where $C^+ = 1$, $C^- = 3$, $K^+ = 1$, and $K^- = \sqrt{3}$.

The original asymptotics are due to Davenport and Heilbronn [?], and the above incorporates improvements by Karim Belabas, Manjul Bhargava, Carl Pomerance, David Roberts, Arul Shankar, Takashi Taniguchi, Jacob Tsimerman and myself [?, ?, ?, ?, ?, ?].

To give a *very* brief overview of the proof, we introduce the lattice of *integral binary cubic forms*, defined by

$$(2) \quad V(\mathbb{Z}) := \{au^3 + bu^2v + cuv^2 + dv^3 : a, b, c, d \in \mathbb{Z}\}.$$

The *discriminant* of $f(u, v) = au^3 + bu^2v + cuv^2 + dv^3 \in V(\mathbb{Z})$ is given by the equation

$$(3) \quad \text{Disc}(f) = b^2c^2 - 4ac^3 - 4b^3d - 27a^2d^2 + 18abcd.$$

The group $\text{GL}_2(\mathbb{Z})$ acts on $V(\mathbb{Z})$ by

$$(4) \quad (\gamma \cdot f)(u, v) = \frac{1}{\det \gamma} f((u, v) \cdot \gamma).$$

A cubic form f is *irreducible* if $f(u, v)$ is irreducible as a polynomial over $\mathbb{Q}$, and *nondegenerate* if $\text{Disc}(f) \neq 0$.

The **Delone-Faddeev correspondence** asserts that these count binary cubic forms:

Theorem 2 ([?, ?, ?]) *There is a canonical, discriminant-preserving bijection between the set of $\text{GL}_2(\mathbb{Z})$ -orbits on $V(\mathbb{Z})$ and the set of isomorphism classes of cubic rings. Under this correspondence, irreducible cubic forms correspond to orders in cubic fields.*

This is one of many such beautiful correspondences. The most classical example is that of binary **quadratic** forms, which parametrize ideal classes in quadratic rings; Gauss discovered a ‘composition law’ which was later seen to correspond to multiplication of ideal classes. More recently, a host of examples were found by Manjul Bhargava in his thesis *Higher Composition Laws* [?, ?, ?, ?], and in further works by Bhargava and his collaborators and students.

So, cubic *orders* correspond to $\mathrm{GL}_2(\mathbb{Z})$ -equivalence classes in some four-dimensional lattice; we have turned an elegant algebraic question into a messy, but hopefully tractable, analytic one. But there is one more twist: we want to count cubic *fields*, which correspond only to the *maximal* orders. Can we describe those in terms of lattice points?

Davenport and Heilbronn proved that the answer is yes, in terms of infinitely many congruence conditions. So our messy question just got messier, but we now have a purely analytic question whose answer is $N_3^\pm(X)$. To make a long story short, the next step is:

- Solve the analytic question, and prove that the answer is the right side of (1)! -

Let’s say we just did that. (Smile and nod.) Now we can ask, *what about arithmetic progressions?* For example, in an analytic number theory course the first big theorem is **the prime number theorem**, which asserts that $\pi(x) \sim \frac{x}{\log x}$, where $\pi(x)$ is the number of primes $\leq x$. The second big theorem is **Dirichlet’s theorem on primes in arithmetic progressions**, which asserts that $\pi(x; q, a) \sim \frac{1}{\phi(q)} \cdot \frac{x}{\log x}$, where $\pi(x; q, a)$ is the number of primes $p \leq x$ and with $p \equiv a \pmod{q}$, where $(a, q) = 1$.

So we investigate the analogue for cubic fields. Here is a table of discriminants of cubic fields K with $0 < \mathrm{Disc}(K) < 2 \cdot 10^6$, arranged by their residue class modulo 7:

Discriminant modulo 7	Count
0	15330
1	17229
2	14327
3	15323
4	17027
5	18058
6	15150

Is this equidistribution with some random noise? Or is something genuinely odd going on? Well, in [?] Taniguchi and I proved the following theorem:

Theorem 3 *Suppose that $(6a, m) = 1$. Then the number of cubic fields K with $0 < \pm \mathrm{Disc}(K) < X$ and $\mathrm{Disc}(K) \equiv a \pmod{m}$ is*

$$(5) \quad N_3^\pm(X; m, a) = \frac{C^\pm}{12\zeta(3)m} \prod_{p|m} \frac{1}{1-p^{-3}} X + K_1(m, a) \frac{4K^\pm}{5\Gamma(2/3)^3} X^{5/6} + O(X^{7/9+\epsilon} m^{8/9}),$$

where

$$(6) \quad K_1(m, a) := \frac{1}{m} \prod_{p|m} \frac{1}{1-p^{-2}} \sum'_{\chi^6=1} \bar{\chi}(a) \frac{L(1/3, \chi^{-2})}{L(5/3, \chi^2)} \prod_{\substack{p|m \\ p \nmid \mathrm{cond}(\chi)}} \frac{1 - \chi(p)^{-2} p^{-4/3}}{1 - \chi(p)^2 p^{-5/3}} \prod_{\substack{p|m \\ p \mid \mathrm{cond}(\chi)}} \frac{\tau_p(\chi_p^2)^3}{p^2}.$$

Here $\tau_p(\chi_p^2) = \sum_{t \in (\mathbb{Z}/p\mathbb{Z})^\times} \chi_p^2(t) e^{2\pi i t/p}$, and the sum is over primitive characters χ to moduli dividing m (including the trivial character modulo 1), such that if we write $\chi = \prod_{p \mid \mathrm{cond}(\chi)} \chi_p$ with each χ_p of conductor p , then each χ_p has exact order 6.

As we wrote before this theorem statement in [?]: “Our general result (Theorem ??) also allows local specifications and is rather complicated to state; the following is a special case.” In contrast to the clean and bare simplicity of the equation above, naturally.

We computed that mess explicitly for the arithmetic progressions $\pmod{5}$ and $\pmod{7}$, so you don’t have to. Here are the results:

Discriminant modulo 5	0	1	2	3	4
Result from (5)	21307	22757	22757	22757	22757
Actual count	21277	22887	22751	22748	22781
Difference	30	130	6	9	24

Discriminant modulo 7	0	1	2	3	4	5	6
Result from (5)	15316	17209	14277	15316	17024	18063	15131
Actual count	15330	17229	14327	15323	17027	18058	15150
Difference	14	20	50	7	3	5	19

Project. *Explain, in geometric terms, what is going on.*

What do I mean by “in geometric terms”? The proofs of Theorems 1 and 3 use the theory of *Shintani zeta functions* – to which involves a lot of complicated machinery – and which, at the end of the day, leave the ‘why’ fairly opaque. It is one thing to come up with a proof, but sometimes it can be a far more difficult thing to understand it.

This project asks you instead to read the work of Bhargava, Shankar, and Tsimerman [?], which gives a proof of Theorem 1 with a somewhat weaker error term, but still obtaining the secondary term. Their proof is still *somewhat* complicated, it involves a lot less heavy machinery, and you should be able to get a good sense of what is going on and at least be able to ‘explain’ some of the data above. For example, why do the (mod 5) and (mod 7) cases behave differently?

There are several natural stopping points to this project:

1. Read [?] together as a group, and understand informally what happens when you introduce arithmetic progressions. For example, if you look at cubic field discriminants in residue classes (mod 11), can you predict whether or not there will be any odd behavior? And if there is, which residue classes will contain the most field discriminants?
2. As an extension of the above, give a proof of Theorem 3, either in whole or in part (e.g. to prime modulus), which avoids Shintani zeta functions and instead adapts the machinery of [?].
3. If you get really excited by this, and want to continue after the workshop, then you can look at *quartic* fields; see [?].

As a warmup, try the following exercise:

Exercise. Let $V(\mathbb{F}_5)$ be the 4-dimensional vector space of binary cubic forms over $\mathbb{F}_5$.

Then $V(\mathbb{F}_5)$ has $5^4 = 625$ elements, each of which has discriminant in $\mathbb{F}_5 = \{0, 1, 2, 3, 4\}$. Determine how many elements of $V(\mathbb{F}_5)$ have each discriminant.

Note that if $x \in V(\mathbb{Z})$ is an integral binary cubic form, then reduction (mod p) and computing the discriminant commute. So, the above exercise should tell you *something* about the distribution of cubic ring discriminants (mod 5).

2 Lattice point counting via elementary and geometric methods

The Gauss Circle Problem and its Extensions

We begin with a discussion of the classical *Gauss Circle Problem*.

For $R > 0$, define

$$N(R) = \#\{(m, n) \in \mathbb{Z}^2 : m^2 + n^2 \leq R^2\}.$$

Thus $N(R)$ is the number of integer lattice points in the closed disk

$$B_R = \{x \in \mathbb{R}^2 : \|x\|_2 \leq R\}.$$

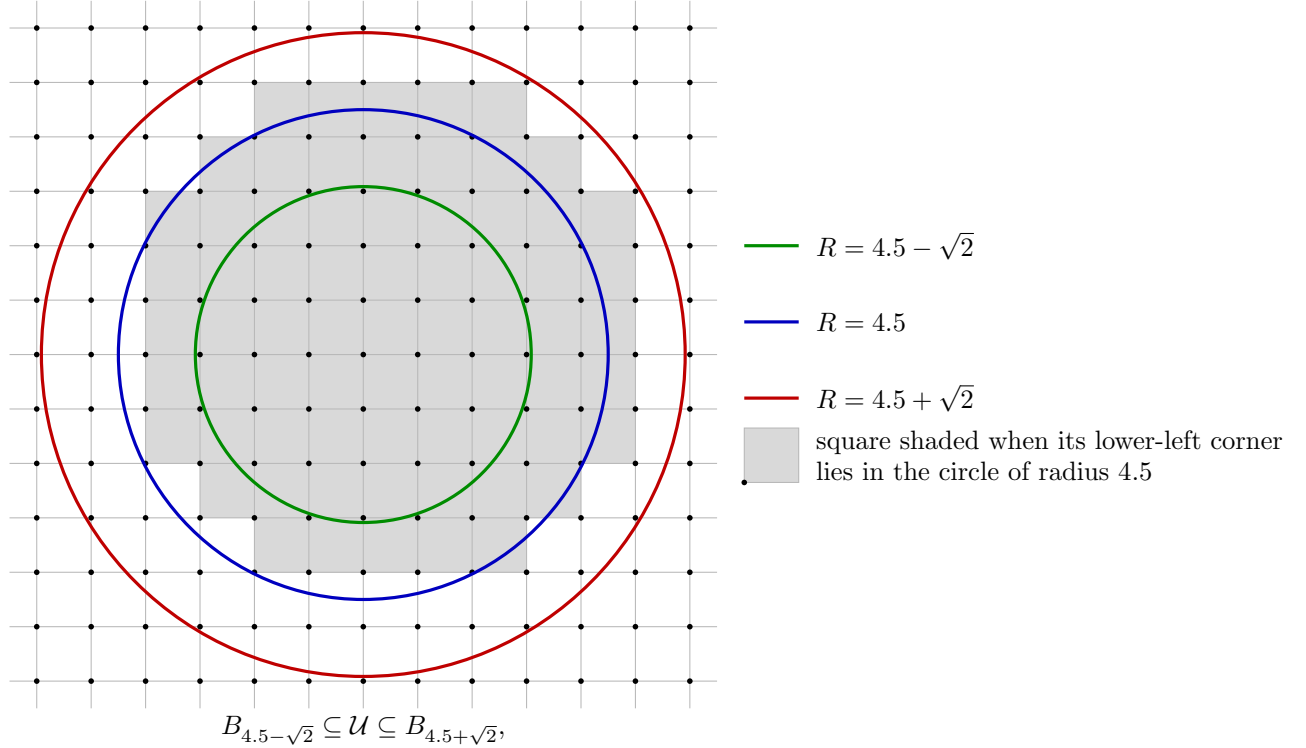
Since the area of B_R is πR^2 , one expects

$$N(R) \sim \pi R^2 \quad (R \rightarrow \infty).$$

We will give an elementary proof that

$$N(R) = \pi R^2 + O(R).$$

The following diagram¹ illustrates the proof in the case $R = 4.5$.



The problem is to count all the lattice points within the middle circle – or, equivalently, compute the area of the squares (shaded in the diagram) whose lower left corner lies within the middle circle.

Draw circles of radius $R \pm \sqrt{2}$, as shown in the diagram. Then, since $\sqrt{2}$ is the diagonal of a unit square,

- The ball of radius $R - \sqrt{2}$ is contained entirely within the shaded area; and,
- The ball of radius $R + \sqrt{2}$ contains entirely the shaded area.

Therefore we have

$$\pi(R - \sqrt{2})^2 \leq N(R) \leq \pi(R + \sqrt{2})^2,$$

and the result follows.

The idea of the proof clearly generalizes, and the following is a natural endpoint:

Theorem 4 (Davenport’s Lemma) *Let $U \subset \mathbb{R}^n$ be a “nice” set. Then, we have*

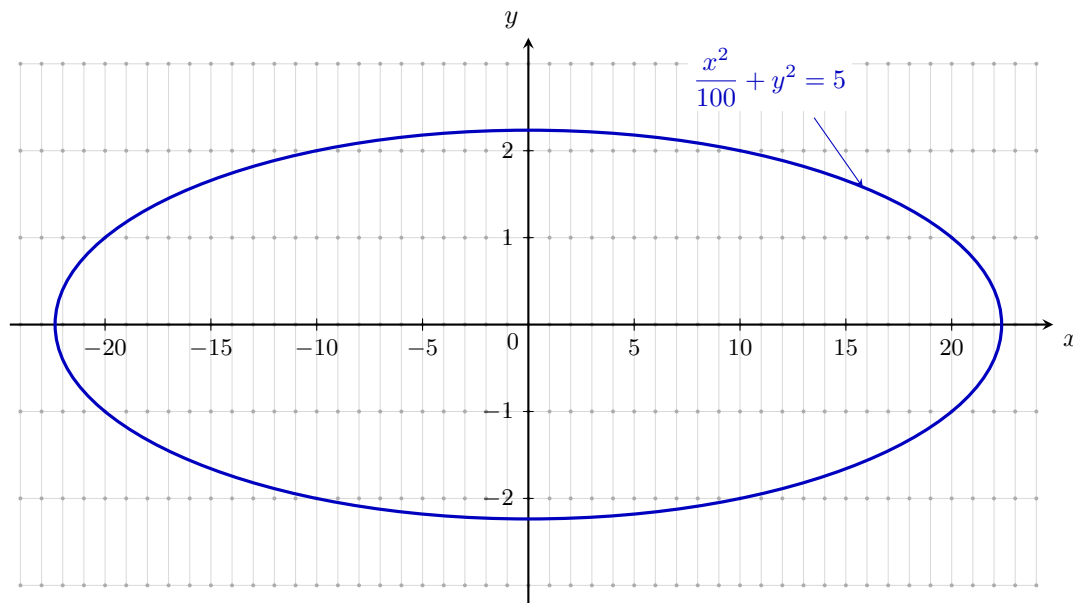
$$\#(U \cap \mathbb{Z}^n) - \text{Vol}(U) \ll \sum_{\pi} \text{Vol}(\pi(U)),$$

where the sum is over all projections π from $\mathbb{R}^n$ to all proper subsets of the coordinate axes.

¹Thanks, ChatGPT!

In practice, *every set you'll come across in arithmetic statistics is nice* (which is not to say it's always trivial to prove this), and we won't worry about the definition here. It is easy to cook up examples of non-nice sets – think of the sort of oddball constructions that show up in real analysis homeworks – but we won't have much reason to care about them.

But much more important are sets which are “nice” in the sense of the above definition, but still problematic. For example, consider the ellipse $\frac{x^2}{n^2} + y^2 = 5$, graphed here with $n = 10$:



The ellipse has much longer projection onto the x -axis than onto the y -axis. Applying Davenport’s Lemma, adapting the proof of the Gauss Circle Problem, or anything similar gives a point count of $5\pi n + O(n)$, which simplifies to the trivial bound of $O(n)$.

That won’t do, and sets like this (but more complicated, and in much higher dimensions) *do* occur and *are* important in arithmetic statistics.

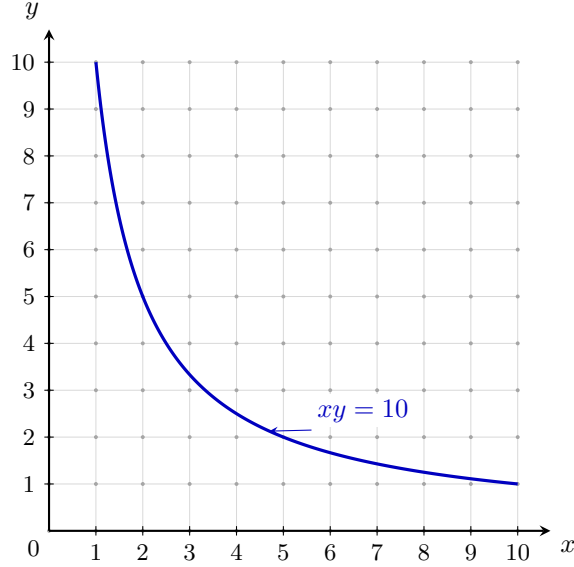
2.1 The hyperbola or divisor problem

Another classical question, of a similar flavor but introducing a new technical obstacle, is the *hyperbola problem*.

We now define² $N(X)$ to be

$$N(X) := \#\{(x, y) \in (\mathbb{Z}^+)^2 : xy \leq X\}.$$

²We’re reusing the same notation, as is common in the arithmetic statistics literature.



In other words, we are counting lattice points under a hyperbola.

Remark 5 If n is a positive integer, the divisor function $d(n)$ counts the number of positive integer divisors of n , or equivalently the number of positive integer solutions to the equation $n = xy$. Therefore, we may also write

$$N(X) = \sum_{n \leq X} d(n),$$

so that our question is equivalent to estimating partial sums of the divisor function.

Following the Gauss circle problem, one might expect $N(X)$ to be well approximated by the area between the coordinate axes and the hyperbola. But one immediately sees a problem! It's an exercise in freshman calculus to see that this area is *infinite*. If we try to apply Davenport's Lemma, the error term there is infinite too.

The region in which we are counting lattice points is *not compact*, and so we can't expect a good result. We leave the solution to this problem as an exercise:

Exercise. Prove that

$$\sum_{n \leq X} d(n) = X \log X + (2\gamma - 1)X + O(X^{1/2}),$$

where

$$\gamma := \lim_{T \rightarrow \infty} \left(\sum_{n \leq T} \frac{1}{n} - \log T \right)$$

is the so-called *Euler-Mascheroni constant*.

This is by no means trivial; thinking for yourself about how to resolve the difficulties we encountered should help you to understand the Bhargava-Shankar-Tsimerman paper.

Lattice point counting in arithmetic statistics: first attempts

Now, let us return to the setting of the Davenport-Heilbronn theorem earlier, and ask for a count of cubic rings R with $0 < |\text{Disc}(R)| < X$. By the Delone-Faddeev correspondence, it is equivalent to count $\text{GL}_2(\mathbb{Z})$ -equivalence classes of binary cubic forms $x \in V(\mathbb{Z})$ with $0 < |\text{Disc}(x)| < X$.

How to handle the $\text{GL}_2(\mathbb{Z})$ -equivalence? The conceptually simplest way is to let U be a *fundamental domain* for the action of $\text{GL}_2(\mathbb{Z})$ on $V(\mathbb{R})$ – i.e., an open set such that every equivalence class in $V(\mathbb{R})$ has

exactly one representative in U . Such a set can be constructed, and can be made to be nice in the sense of Davenport's Lemma, but there is a problem, illustrated by the following exercise:

Exercise. Prove that for any real numbers $a, X > 0$, there exists $\epsilon_{a,X} \in (0, 1)$ for which

$$0 < |\text{Disc}(au^3 + \epsilon u^2v + 0uv^2 + 0v^3)| < X.$$

If we fix X , and let W be the set of all $(a, \epsilon_{a,X}, 0, 0) = au^3 + \epsilon u^2v + 0uv^2 + 0v^3$ satisfying the above, then W is not compact, and its projection onto the u^3 axis is all of $(0, \infty)$. So, if one tried to apply Davenport's Lemma, one would obtain an *infinite* error term.

It turns out that there's no way to avoid this flavor of problem; the fundamental domain U cannot be chosen compact. But there is reason for hope: any *integer* lattice point $(a, 0, 0, 0)$ necessarily has discriminant 0. So perhaps we can figure out where to cut off the fundamental domain....?

Bhargava's averaging method

We now jump forward to the centerpiece of Bhargava's averaging method. We will follow the proof in [?] of the Davenport-Heilbronn, but exactly the same technique works for a host of other representations.

Here is the key formula – (16) from [?] – which I have shamelessly copy and pasted from their LaTeX source:

$$(7) \quad N(V(\mathbb{Z})^{(i)}; X) = \frac{\int_{v \in B \cap V(\mathbb{R})^{(i)}} \#\{x \in \mathcal{F}v \cap V(\mathbb{Z})^{\text{irr}} : |\text{Disc}(x)| < X\} |\text{Disc}(v)|^{-1} dv}{n_i \cdot \int_{v \in B \cap V(\mathbb{R})^{(i)}} |\text{Disc}(v)|^{-1} dv}.$$

It is a doozy, but wrap your head around this and you've basically won the ballgame. What is all of this?

- $V(\mathbb{Z})$ is the lattice of integral binary cubic forms, and $V(\mathbb{R})$ is the real vector space of real binary cubic forms. $V(\mathbb{Z})^{\text{irr}}$ is the subset of *irreducible* binary cubic forms – i.e., those which don't factor over $\mathbb{Z}$, or equivalently whose corresponding cubic rings are integral domains. The subscript (i) corresponds to either of the two *nonsingular* $G(\mathbb{R})$ -orbits of binary cubic forms: those which factor into distinct linear factors over $\mathbb{R}$, and have positive discriminant; and those which have an irreducible quadratic factor over $\mathbb{R}$, and have negative discriminant.
- The counting function $N(V(\mathbb{Z})^{(i)}; X)$ counts $\text{GL}_2(\mathbb{Z})$ -orbits of irreducible binary cubic forms x , such that x is in the desired $G(\mathbb{R})$ -orbit, with $0 < |\text{Disc}(x)| < X$.
- B is a compact subset of $V^{(i)}(\mathbb{R})$, which *we can take to be almost anything*. All we require, for technical reasons, is that it be $\text{SO}_2(\mathbb{R})$ -invariant.
- $\mathcal{F}$ is a fundamental domain for the action of $\text{GL}_2(\mathbb{Z})$ on $\text{GL}_2(\mathbb{R})$ by left multiplication. Although the formula works with any fundamental domain, we will want to choose a specific one. Again copying from [?]: $\mathcal{F}$ may be expressed in the form $\mathcal{F} = \{nak\lambda : n \in N'(a), a \in A', k \in K, \lambda \in \Lambda\}$, where

$$(8) \quad N'(a) = \left\{ \begin{pmatrix} 1 & \\ n & 1 \end{pmatrix} : n \in \nu(a) \right\}, \quad A' = \left\{ \begin{pmatrix} t^{-1} & \\ & t \end{pmatrix} : t \geq \sqrt[4]{3}/\sqrt{2} \right\}, \quad \Lambda = \left\{ \begin{pmatrix} \lambda & \\ & \lambda \end{pmatrix} : \lambda > 0 \right\},$$

and K is as usual the (compact) real special orthogonal group $\text{SO}_2(\mathbb{R})$; here $\nu(a)$ is the union of either one or two subintervals of $[-\frac{1}{2}, \frac{1}{2}]$ depending only on the value of $a \in A'$. Furthermore, if a is such that $t \geq 1$, then $\nu(a) = [-\frac{1}{2}, \frac{1}{2}]$.

- n_i is the cardinality of the stabilizer in $\text{GL}_2(\mathbb{R})$ of any $v \in V(\mathbb{R})^{(i)}$. This turns out to be 6 for positive discriminant forms, and 2 for the negative discriminant.

The key is that the simpler formula

$$N(V(\mathbb{Z})^{(i)}; X) = \frac{1}{n_i} \#\{x \in \mathcal{F}v \cap V(\mathbb{Z})^{\text{irr}} : |\text{Disc}(x)| < X\}$$

is true for every v , and this is essentially a tautology: stare at it deeply until you convince yourself that it's obvious. (Or "obvious", as the case may be.)

Believe that, and it's immediate that one can integrate over v , and throw in that factor of $|\text{Disc}(v)|^{-1}$ (or whatever else you please), and you still get a true formula.

So why write down (7)? Because we can change variables! Proposition 23 of [?] is the formula

$$\int_{g \in \text{GL}_2(\mathbb{R})} f(g \cdot v_i) dg = \frac{1}{2\pi} \int_{v \in \text{GL}_2(\mathbb{R}) \cdot v_i} f(v) |\text{Disc}(v)|^{-1} dv = \frac{n_i}{2\pi} \int_{v \in V_{\mathbb{R}}^{(i)}} f(v) |\text{Disc}(v)|^{-1} dv,$$

valid for any $v_i \in V(\mathbb{R})^{(i)}$ and essentially any function f supported on $V(\mathbb{R})^{(i)}$. This follows from the Jacobian change of variables formula from $\text{GL}_2(\mathbb{R})$ to $V(\mathbb{R})$, using the change of variables $g \mapsto gv_i$, which is locally a homeomorphism. This is *not* obvious, there is a serious computation to be done, but it should not be surprising that a formula like this is true. We thus obtain

$$(9) \quad N(V^{(i)}(\mathbb{Z}); X) = C_i \int_{g \in \mathcal{F}} \#\{x \in V(\mathbb{Z})^{\text{irr}} \cap gB \cap V(\mathbb{R})^{(i)} : |\text{Disc}(x)| < X\} dg.$$

And this is nice. Why?

- The change of variables has moved the 'cusp' from the subset of $V(\mathbb{R})$ to the integral over $\mathcal{F}$. B is something nice – basically a ball! We now need to count lattice points in a ball, translated by some linear transformation, and then 'cut off' by the condition $|\text{Disc}(x)| < X$. Davenport's Lemma works perfectly for this!
- The irreducibility condition looks scary, but it actually *helps* us. The key is the t variable occurring in the decomposition of $\mathcal{F}$ above. When t is small, the action of g doesn't do anything too crazy to B , and most of the lattice points are irreducible. See the exercise below, which asks you to prove a special case of this.

When t is large, the action of g skews B a lot, and this is the exact scenario when we get worse error terms from Davenport's Lemma. But if t is big enough, you can check that the a coefficient of an element of gB must satisfy $|a| < 1$. But if a is an integer, that means it's zero, which in turn means any corresponding form is reducible. So we can stop counting here!

Exercise. There are $O(B^4)$ integral binary cubic forms (a, b, c, d) with $|a|, |b|, |c|, |d| \leq B$. Prove any nontrivial bound on the number of *reducible* such forms.

3 Sato-Shintani zeta functions and their application

In this lecture, we introduce the zeta function theory of Sato and Shintani, and say what it's good for.

Warmup: the Gauss Circle Problem

As a warmup, let's revisit the Gauss circle problem. Define

$$Z(s) := \sum_{\substack{n, m \in \mathbb{Z} \\ (n, m) \neq (0, 0)}} (n^2 + m^2)^{-s},$$

which makes a Dirichlet series out of our counting question. It turns out that $Z(s)$ belongs to two families of zeta functions simultaneously! We have $Z(s) = 4\zeta_{\mathbb{Z}[i]}(s)$, where $\zeta_{\mathbb{Z}[i]}(s)$ is the *Dedekind zeta function*

associated to the number ring $\mathbb{Z}[i]$. Alternatively, $Z(s)$ is an *Epstein zeta function*, enumerating the values taken by a positive definite quadratic form.

Therefore, $Z(s)$ is known to be a good zeta function. It has a meromorphic continuation to all of $\mathbb{C}$, holomorphic apart from a simple pole at $s = 1$. And it enjoys a functional equation, which allows us to predict its growth outside the region of absolute convergence. (There is also an Euler product, but we won't need it, and unlike the first two properties the Euler product goes away if we tweak the setup slightly.)

By Perron's formula, we have

$$N(R) - 1 = \#\{(n, m) \in \mathbb{Z}^2, (n, m) \neq (0, 0), n^2 + m^2 \leq R^2\} = \frac{1}{2\pi i} \int_{2-i\infty}^{2+i\infty} Z(s)(R^2)^s \frac{ds}{s}.$$

By “standard methods” we then have

$$N(R) = \text{Res}_{s=1} \left(Z(s) \frac{(R^2)^s}{s} \right) + O(R^{2/3}) = \pi R^2 + O(R^{2/3}),$$

and we recover the solution to the Gauss circle problem – with a better error term to boot.

Shintani's zeta function

In [?], Sato and Shintani developed a theory of *zeta functions associated to prehomogeneous vector spaces*. The space V of binary cubic forms, together with the action of GL_2 , is prehomogeneous, meaning that *considered over $\mathbb{C}$, the action has a Zariski open orbit*. In particular, a cubic form has discriminant 0 if and only if it has a multiple root, and –

Exercise. Let f_1 and f_2 be any two complex binary cubic forms, each of which factors over $\mathbb{C}$ into distinct linear factors. Prove that there exists $g \in \text{GL}_2(\mathbb{C})$ with $g \circ f_1 = f_2$.

(Hint: Factor both forms, and finding g sending each linear factor in f_1 to its corresponding linear factor in f_2 . Since the factors can be ordered arbitrarily, there are six g satisfying the desired conclusion.

Sato and Shintani proved that such zeta functions enjoy analytic continuation and a functional equation. In the case of binary cubic forms, Shintani [?] further determined the locations of the poles with their residues, and computed the functional equation explicitly. Define the *Shintani zeta functions*

$$(10) \quad \xi^\pm(s) = \sum_n a^\pm(n) n^{-s} := \sum_{\substack{x \in \text{GL}_2(\mathbb{Z}) \backslash V(\mathbb{Z}) \\ \pm \text{Disc}(x) > 0}} \frac{1}{|\text{Stab}(x)|} |\text{Disc}(x)|^{-s},$$

and note that by the Delone-Faddeev correspondence we have

$$(11) \quad \xi^\pm(s) = \sum_{\pm \text{Disc}(R) > 0} \frac{1}{|\text{Aut}(R)|} |\text{Disc}(R)|^{-s},$$

where the sum is over cubic rings. The factors of $|\text{Stab}(x)|$ and $|\text{Aut}(R)|$ are well understood, and bounded above by the same six that you saw if you worked out the exercise above – if you are not an expert in this subject, you can basically pretend they're not there.

Shintani established that these are holomorphic apart from simple poles at $s = 1$ and $s = 5/6$, and Landau's method yields that

$$(12) \quad \sum_{0 < \pm \text{Disc}(R) < X} \frac{1}{|\text{Aut}(R)|} = \text{Res}_{s=1}(\xi^\pm(s))X + \text{Res}_{s=5/6}(\xi^\pm(s)) \frac{X^{5/6}}{5/6} + O(X^{3/5}).$$

So, if you're willing to take all this machinery for granted, and look up the residues from Shintani's paper without trying to read the proofs too carefully, we're essentially done with our counting problem!

Moreover, the factor of $\frac{1}{|\text{Aut}(R)|}$ is well understood (see Proposition ??), and can be removed without too much effort.

Maximality and the sieve

We have, so far, been ignoring one important technical condition. The methods above count all cubic rings $R \neq 0$; Bhargava's method imposes an additional irreducibility constraint, whereas Shintani's method allows you to subtract the reducible rings afterwards.

But the cubic fields correspond to the *maximal* cubic rings, and we have yet to say anything about maximality. In [?], Davenport and Heilbronn proved that a cubic ring R is maximal, if and only if f satisfies the following condition for every prime p :

- the cubic form f is not a multiple of p ; and
- there is no $\mathrm{GL}_2(\mathbb{Z})$ -transformation of $f(u, v) = au^3 + bu^2v + cuv^2 + dv^3$ such that a is a multiple of p^2 and b is a multiple of p .

Exercise. Suppose that f is an integral form failing to meet one of the two conditions above, for a given prime p . Find $g \in \mathrm{GL}_2(\mathbb{Q})$, such that $g \cdot f$ is also an integral form, and $\det(g)$ is a negative power of p .

Can we count elements of lattices satisfying infinitely many congruence conditions. Well, no. At least not directly. But our methods to translate directly to *finitely* many congruence conditions:

Definition 6 Let q be a squarefree integer, and f an integral binary cubic form. We say that f is *nonmaximal* at q if it fails the Davenport-Heilbronn condition described above, for **every** prime dividing q .

We then have, by inclusion-exclusion, that

$$(13) \quad \sum_{\substack{0 < \pm \mathrm{Disc}(R) < X \\ R \text{ maximal}}} \frac{1}{|\mathrm{Aut}(R)|} = \sum_{q=1}^{\infty} \mu(q) \sum_{\substack{0 < \pm \mathrm{Disc}(R) < X \\ R \text{ nonmaximal at } q}} \frac{1}{|\mathrm{Aut}(R)|}.$$

We then divide the right hand side into ‘small q ’ – $q < X^\theta$ for some $\theta > 0$ to be determined, say, and ‘large q ’ – the rest.

For large q we cannot hope for any of the tools described here to directly work. But, luckily, we have the

Tail Estimate. We have

$$\sum_{\substack{0 < \pm \mathrm{Disc}(R) < X \\ R \text{ nonmaximal at } q}} 1 \ll \frac{X}{q^{2-\epsilon}}.$$

This is not too difficult to prove, but we omit the proof for now. But note, importantly, that this can be proved for ‘special reasons’, and analogues *can't* be proved in many situations where they are expected to hold, and would be *very* useful if they could be proved. See [?] for a striking example of a nice theorem where a tail estimate is the key technical input – and where the authors resort to a rather crazy-looking construction to prove it!

We turn then to the small q , where we introduce the Shintani zeta functions corresponding to the ‘nonmaximal at q ’ condition. Let $\Phi_{q^2} : V(\mathbb{Z}/q^2\mathbb{Z}) \rightarrow \{0, 1\}$ be the characteristic function of forms which are nonmaximal at q , and we also write Φ_{q^2} for its lift to a function $V(\mathbb{Z}) \rightarrow \{0, 1\}$.

We then define

$$(14) \quad \xi^\pm(s, \Phi_{q^2}) = \sum_n a^\pm(\Phi_{q^2}, n) n^{-s} := \sum_{\substack{x \in \mathrm{GL}_2(\mathbb{Z}) \backslash V(\mathbb{Z}) \\ \pm \mathrm{Disc}(x) > 0}} \frac{1}{|\mathrm{Stab}(x)|} \Phi_{q^2}(x) |\mathrm{Disc}(x)|^{-s},$$

and Landau's method yields

$$(15) \quad \sum_{\substack{0 < \pm \mathrm{Disc}(R) < X \\ \text{nonmaximal at } q}} \frac{1}{|\mathrm{Aut}(R)|} = \mathrm{Res}_{s=1}(\xi^\pm(s))X + \mathrm{Res}_{s=5/6}(\xi^\pm(s)) \frac{X^{5/6}}{5/6} + O_q(X^{3/5}).$$

If we obtain a more explicit error term of the shape $O(q^\alpha X^{3/5})$, then our total error is

$$\ll X^{3/5} \sum_{q < X^\theta} q^\alpha + \sum_{q \geq X^\theta} \frac{X}{q^{2-\epsilon}} \ll X^{3/5} X^{\theta \cdot (1+\alpha)} + X^{1-\theta+\epsilon}.$$

(This assumes that $\alpha > -1$. If $\alpha = -1$, the first error term is $O(X^{3/5} \log X)$, and if $\alpha < -1$ it is $O(X^{3/5})$.) Given α , we choose θ to equalize the two error terms, and we are done!

Landau's method

We conclude this section by saying something about Landau's method, which so far we've treated as a black box. The starting point is the following:

Proposition 7 *Let $L(s) := \sum_{n \geq 1} a(n)n^{-s}$ be a Dirichlet series. Then, we have*

$$(16) \quad \sum_{n < X} a(n) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} L(s) X^s \frac{ds}{s},$$

valid for any c for which the Dirichlet series defining $L(s)$ converges absolutely.

Note that this statement doesn't depend on any sort of analytic continuation or functional equation or the like. To prove it, replace $L(s)$ by its definition, switch the order of summation and integration, and use the identity

$$(17) \quad \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} x^s \frac{ds}{s} = \begin{cases} 1, & \text{if } x > 1, \\ 0, & \text{if } 0 < x < 1, \end{cases}$$

which is an exercise (not trivial, but not terribly difficult either) in complex analysis. One must worry about convergence, but there are no problems here. We assume that X is not an integer to avoid an additional $\frac{1}{2}a(X)$ term on the left.

How might one use this? Let's try applying it to the Gauss circle problem; we work *formally* and leave any convergence questions for later. As before we have

$$N(X) - 1 = \frac{1}{2\pi i} \int_{2-i\infty}^{2+i\infty} Z(s) X^s \frac{ds}{s}.$$

Shift the contour to the line $\Re(s) = \frac{1}{2}$, passing a simple pole at $s = 1$, and computing the residue we conclude that

$$(18) \quad \sum_{n < X} a(n) = \pi X + E(X),$$

for an error term E satisfying

$$(19) \quad |E(X)| \leq X^{1/2} \cdot \frac{1}{2\pi} \int_{1/2-i\infty}^{1/2+i\infty} |\zeta(s)|^2 \left| \frac{ds}{s} \right|.$$

Since the integrand is independent of X , as long as it is absolutely convergent we have $E(X) \ll X^{1/2}$.

We notice two things about this computation:

1. This relies on analytic continuation of $Z(s)$ to $\Re(s) = 1/2$, where it is not defined by its Dirichlet series. At the heart of the usual proof of analytic continuation is an application of Poisson summation.
2. Unfortunately, the convergence issues are serious here. The integral in (19) does *not* converge, and the above cannot be rigorously justified. So, one must refine the method so as to allow a rigorous proof.

Before describing Landau's method, which gives a way of handling the convergence issues, we present a sample result. The following variation is due to Lowry-Duda, Taniguchi, and the author [?]:

Theorem 8 *Let $\phi(s) = \sum_n a(n)\lambda_n^{-s}$ be a zeta function with nonnegative coefficients, absolutely convergent for $\Re(s) > 1$, enjoying an analytic continuation to $\mathbb{C}$ which is holomorphic away from a simple pole at $s = 1$, and with a 'well behaved' functional equation of degree d relating $\phi(s)$ to $\widehat{\phi}(1-s)$ for a 'dual zeta function' $\widehat{\phi}(s) = \sum_n b(n)\mu_n^{-s}$.*

Then, excluding a couple of uncommon special cases, for $X \geq 2$ we have

$$(20) \quad \sum_{\lambda_n < X} a(n) = \text{Res}_{s=1}(\phi(s))X + O\left(X^{\frac{d-1}{d+1}}\delta_1^{\frac{d-1}{d+1}}\widehat{\delta}_1^{\frac{2}{d+1}}\right),$$

where

$$\begin{aligned} \delta_1 &= \text{Res}_{s=1}(\phi(s)), \\ \widehat{\delta}_1 &= \sup_Z \frac{1}{Z} \sum_{\mu_n < Z} |b(n)|. \end{aligned}$$

The implied constant depends on the functional equation, but does not depend further on $\phi(s)$ or the $a(n)$.

The statement generalizes further – to allow higher order poles, to allow poles at other locations (such as $s = \frac{5}{6}$, as in the Shintani zeta function), to allow negative coefficients, and so forth. Unfortunately the most general version of this result, although not so complicated theoretically, is rather complicated to state.

Note that the error term may be written simply as $O(X^{\frac{d-1}{d+1}})$, where the implied constant depends on ϕ . In typical examples, ϕ is fixed and X is allowed to increase, and the precise ϕ -dependence is not of much interest. In this case, our work reduces to results obtained by Landau [?, ?] and Chandrasekharan and Narasimhan [?]. The novelty of [?] was to carefully track the ϕ -dependence.

We will apply this to the Shintani zeta functions of the form defined in (??), and the uniformity will allow us to track the dependence on Φ_m . From the standpoint of Landau's method, the zeta function dual to $\xi^\pm(s, \Phi_m)$ is $m^{4s}\xi^\pm(s, \widehat{\Phi_m})$, where we define $\widehat{\Phi_m} : V^*(\mathbb{Z}/m\mathbb{Z}) \rightarrow \mathbb{C}$ by the usual Fourier transform formula

$$(21) \quad \widehat{\Phi_m}(x) := \frac{1}{m^4} \sum_{y \in V(\mathbb{Z}/m\mathbb{Z})} \Phi_m(y) \exp\left(2\pi i \cdot \frac{[x, y]}{m}\right),$$

and lift $\widehat{\Phi_m}$ to a function on $V^*(\mathbb{Z})$, which we identify with $V(\mathbb{Z})$. (There are some mild technicalities in this identification, which we sweep under the rug.)

Note the factor of m^{4s} ! This comes from the functional equation, but if we include it in the definition of the zeta function – as we will do here – then the functional equation will be the same for all Φ_m simultaneously, so that we can use our 'Uniform Landau' Theorem 8. We obtain the following:

Theorem 9 *As described above, given a zeta function pair*

$$\xi^\pm(s, \Phi_m) := \sum_n a^\pm(\Phi_m, n)n^{-s}, \quad \xi^\pm(s, \widehat{\Phi_m}) := \sum_n a^\pm(\widehat{\Phi_m}, n)n^{-s}.$$

Let

$$(22) \quad \delta_1 = \delta_1(\Phi_m) := \text{Res}_{s=1} \xi^\pm(s, \Phi_m)$$

and

$$(23) \quad \widehat{\delta}_1 = \widehat{\delta}_1(\Phi_m) := m^4 \cdot \sup_N \frac{1}{N} \sum_{\alpha \in \{\pm\}} \sum_{n < N} a^\alpha(|\widehat{\Phi_m}|, n),$$

Assuming two technical conditions which we omit here, we have

$$(24) \quad N^\pm(X, \Phi_m) := \sum_{n < X} a^\pm(\Phi_m, n) = \sum_{\sigma \in \{1, \frac{5}{6}\}} \frac{X^\sigma}{\sigma} \cdot \text{Res}_{s=\sigma} \xi^\pm(s, \Phi_m) + O\left(X^{3/5} \delta_1^{3/5} (\widehat{\delta}_1)^{2/5}\right).$$

We will end up summing the error term over many different Φ_m . In practice we will have something like $\delta_1(\Phi_m) \ll \frac{1}{m}$ for many Φ_m of interest – and naturally we are happy to see negative exponents of m in the error term! What is clear from this formula is that we want to bound $\widehat{\delta}_1$ – and hence $\widehat{\Phi_m}$ – as much as we can.

Idea of the proof of uniform Landau. To even give a reasonable sketch of the proof would take us too far afield, but we will at least give the idea. As explained above, we have

$$\sum_{n < X} a^\pm(\Phi_m, n) = \frac{1}{2\pi i} \int_{2-i\infty}^{2+i\infty} \xi^\pm(s, \Phi_m) X^s \frac{ds}{s}$$

and morally the idea is to shift the contour to the left of the critical strip, picking up polar contributions at $s = 1$ and $s = 5/6$, and using the functional equation of the zeta function to estimate and bound the resulting integral.

Unfortunately, the zeta function *grows* along vertical lines to the left of the critical strip. This also implies that known bounds grow *within* the critical strip.

To remedy the convergence issues, we consider the following variation of Perron's formula, based on *Riesz means*, which holds for each positive integer k :

$$(25) \quad N(X, \Phi_m, k) := \frac{1}{k!} \sum_{n < X} a(\Phi_m, n) \left(1 - \frac{n}{X}\right)^k = \frac{1}{2\pi i} \int_{2-i\infty}^{2+i\infty} \xi(s, \Phi_m) \frac{X^s}{s(s+1)\cdots(s+k)} ds.$$

As the growth of ξ is *polynomial* along vertical lines, by taking k large enough we obtain convergence of the integral if we shift the line to $\Re(s) = c$, for $c \in [0, 1]$, or for c slightly smaller than 0. The above strategy then works as it was earlier described.

We could simply declare victory for a variant of our cubic field counting problem, where we count each field with $|\text{Disc}(K)| < X$ with weight $(1 - |\text{Disc}(K)|/X)^k$. This is an example of a general principle in number theory, that incorporating smooth weights often leads to simpler analysis and/or better error terms. (At heart, the general principle comes from Fourier analysis: the smoother the function, the more rapidly its Fourier transform decays.) See Shankar, Södergren, and Templier [?] for an example where the authors ‘declare victory’ with a different smoothing function, and then obtain striking results on the central values of the associated Dedekind zeta functions.

Instead, we recover the unsmoothed estimate, up to an error term. Define the k th finite difference operator Δ_y^k by

$$(26) \quad \Delta_y^k F(X) = \sum_{\nu=0}^k (-1)^{k-\nu} \binom{k}{\nu} F(X + \nu y),$$

and then in (25) we have

$$(27) \quad N(X, \Phi_m) = y^{-k} \Delta_y^k (X^k N(X, \Phi_m, k)) + O\left(\sum_{X \leq n \leq X+ky} |a(\Phi_m, n)|\right).$$

Provided that one can also analyze the finite differences of the shifted integral in (25), we are able to obtain the theorem.

4 Incorporating Fourier analysis to improve error terms

We've now seen two methods for counting lattice points. Suppose we want the best error terms? Then, one way or another, we have to use techniques from Fourier analysis.

The basic tool is the

Poisson Summation Formula. Suppose that f is a 'reasonably nice' function on $\mathbb{R}^n$. Then, we have

$$(28) \quad \sum_{x \in \mathbb{Z}^n} f(x) = \sum_{y \in \mathbb{Z}^n} \widehat{f}(y),$$

where

$$\widehat{f}(y) := \int_{\mathbb{R}^n} f(x) e^{-2\pi i x \cdot y} dx.$$

In particular, the formula holds true for f in the *Schwartz class*: infinitely differentiable, such that f and all of its derivatives decay more rapidly than $|x|^{-A}$ as $|x| \rightarrow \infty$, for any A .

Let us try applying this to the Gauss circle problem; we will recover the same error term that we got via the zeta function/Landau machinery. Write g for the characteristic function of the unit ball B_1 , and $f(x) := g(x/R)$, so that the left side of (31) is the lattice point count we're looking for. This f is not a Schwartz function, but let us ignore that for now, as a more serious difficulty will soon arise.

Then $\widehat{f}(0)$ is simply the area of B_R , or πR^2 , the expected main term. For the remaining terms, the Fourier transform $\widehat{g}$ can be computed in terms of something called a *Bessel function*, and it satisfies

$$|\widehat{g}(y)| \ll (1 + |y|)^{-3/2}.$$

It can be then easily checked that

$$|\widehat{f}(y)| \ll R^2 (1 + R|y|)^{-3/2},$$

so that we obtain an error term bounded by

$$\sum_{(0,0) \neq y \in \mathbb{Z}^2} R^2 (1 + R|y|)^{-3/2}.$$

This does not converge, and hence we obtain an infinite error term.

To remedy this situation, we borrow the previous strategy of proving upper and lower bounds. We majorize f by a function of the form

$$(29) \quad F := g\left(\frac{\cdot}{R+c}\right) * \phi_c,$$

where $c < 1$ is a small positive real number, ϕ is a Schwartz function of total mass 1 supported on the unit ball³, $\phi_c(x) := c^2 \phi(x/c)$ is this function normalized to be supported on the ball of radius c , and the *convolution* $*$ is defined by

$$(f * g)(x) := \int f(y) g(x - y) dy.$$

Essentially, we start with the characteristic function of the ball of radius $R + c$, and then 'smooth it out' by averaging over its neighbors in a ball of radius c . This F will be supported within the ball of radius $R + 2c$, will be identically 1 within the ball of radius R , and will be smooth on the small transition region between. In particular, F will majorize f . We then have

$$\sum_{y \in \mathbb{Z}^2} \widehat{F}(y) = (R + c)^2 \sum_{y \in \mathbb{Z}^2} \widehat{g}((R + c)y) \widehat{\phi}_c(y).$$

³It is not obvious that such a function exists, but it is 'well known'.

The main term, coming from $y = 0$, is $\pi(R + c)^2$. We will now give a quick and dirty bound on the error term; as an exercise, turn all of this into a rigorous proof.

We have $\widehat{\phi}_c(y) = c^2 \widehat{\phi}(cy)$, limiting the effective support on the right to $|y| \ll c^{-1}$. For any $T \ll c^{-1}$, there are $O(T^2)$ choices of $y \in \mathbb{Z}^2$ with $|y| \asymp T$, and for each of them we have $|\widehat{g}((R + c)y)| \ll (TR)^{-3/2}$ so that all these y contribute $O(T^{1/2}R^{1/2})$ to the error term. Summing over $T \ll c^{-1}$, where T ranges over the power of 2, we obtain an error term of $O(c^{-1/2}R^{1/2})$.

Putting this all together, we obtain

$$N(R) \leq \pi(R + c)^2 + O(c^{-1/2}R^{3/2}) = \pi R^2 + O(Rc) + O(c^{-1/2}R^{1/2}),$$

and choose $c = R^{-1/3}$ to obtain

$$N(R) \leq \pi R^2 + O(R^{2/3}).$$

We now repeat the argument, replacing the majorant (29) by the minorant

$$(30) \quad F := g\left(\frac{\cdot}{R - c}\right) * \phi_c,$$

to obtain

$$N(R) \geq \pi R^2 + O(R^{2/3}),$$

and hence an asymptotic with error term $O(R^{2/3})$.

Poisson summation with congruence conditions

Given $\Phi_m : (\mathbb{Z}/m)^n \rightarrow \mathbb{C}$, suppose that we want to estimate

$$\sum_{x \in U \cap \mathbb{Z}^n} \Phi_m(x).$$

One can do well by dividing $\mathbb{Z}^n$ into m^n residue classes $(\bmod m)$ and applying Davenport's Lemma to each. However, one does better by applying the

Poisson Summation Formula with Congruence Conditions. Suppose that f is a ‘reasonably nice’ function on $\mathbb{R}^n$. Then, we have

$$(31) \quad \sum_{x \in \mathbb{Z}^n} f(x) \Phi_m(x) = \sum_{y \in \mathbb{Z}^n} \widehat{f}\left(\frac{y}{m}\right) \widehat{\Phi}_m(y),$$

where

$$\widehat{\Phi}_m(y) := m^{-n} \sum_{x \in (\mathbb{Z}/m)^n} \Phi_m(x) e^{-2\pi i x \cdot y / m}.$$

is the Fourier transform of Φ_m , but over $(\mathbb{Z}/m)^n$ rather than $\mathbb{R}^n$.

(To do: check the sign normalization)

So, not only does our method still work, but it allows us to obtain additional savings in our final results by studying $\widehat{\Phi}_m$. We now see the relevance of Proposition ?? and other results of the same flavor! Bounds on $|\widehat{\Phi}_m|$, whether pointwise or on average, can be thought of as a type of equidistribution result for Φ_m – and it is natural to hope for such an equidistribution result to be reflected in better error terms for lattice point counting.

So how does Fourier analysis show up in the arithmetic statistics literature?

- **Elementary lattice point counting methods.** Fourier analysis has been used in combination with fairly elementary methods; see, for example, work of Belabas and Fouvry [?] which obtained a ‘level of distribution’ for binary cubic forms, and work of myself with Taniguchi [?] which further developed this approach.

- **Bhargava’s averaging method.** For the most part, papers using this method don’t apply a lot of Fourier analysis. However, one nice – and elementary! – use appears in [?, Section 9.4], where they prove that

$$\sum_{\mathcal{B} \cap V_{\mathbb{Z}}} \Phi_m(v) = \widehat{\Phi_m}(0) \cdot \text{Vol}(\mathcal{B}) + O(m^{3+\epsilon}),$$

for V the 4-dimensional lattice of binary cubic forms, and Φ_m a particular function of the shape above.

A different use comes in forthcoming work of Anderson, Bhargava, and myself [?], where we incorporate Fourier analysis into the heart of Bhargava’s method in (7) and (9). The technique should adapt widely, and we expect to be able to obtain quantitative improvements in a variety of results.

- **Shintani zeta functions.** Here, the Poisson summation formula is baked into the machinery, as it’s the key to proving the functional equation. (This is also true for the simpler case of the Riemann zeta function.) So, any argument involving Shintani zeta functions is implicitly relying on Fourier analysis. Moreover, Landau’s method relies on the formulas (16) and (17); although these can be proved without any knowledge of Fourier analysis, they are best understood in terms of the *Mellin transform*, which is related to the Fourier transform by a change of variable. Moreover, once the machinery has been unraveled, the variation (25) illustrates a basic principle of Fourier analysis: that the smoother a function is, the faster its Fourier transform decays.